

RITHISH R

📞 +91 6382997458 | ✉️ rithishkumarravi2@gmail.com | 🌐 rith-19 | 📺 Rithish R |

EDUCATION

Madras University

M.Sc Computer Science

Chennai, TN

Sept 2022 to Apr 2024

Guru Nanak College

B.Sc Computer Science

Chennai, TN

May 2019 to July 2022

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL

AI / ML Frameworks & Tooling: PyTorch, TensorFlow, Hugging Face Transformers, PEFT, Accelerate, LangChain, Ultralytics YOLO, OpenCV

LLMs & Generative AI: Open-source LLMs and Code Generation Models, Prompt Engineering, RAG Pipelines, Multi-Agent Systems, Groq APIs

ML Training & Optimization: Supervised Fine-Tuning (SFT), Knowledge Distillation, RLHF (PPO / DPO), Structured Pruning (WANDA), Quantization, GPU Optimization

Computer Vision & Video Analytics: Object Detection, Image Segmentation (SAM), Face Recognition, Person/Vehicle ReID, Automated License Plate Recognition (ALPR), Real-time Video Processing

Backend, Data & MLOps: Supabase, Prisma ORM, PostgreSQL, REST API Design, Git, Pandas, NumPy, Dataset Synthesis

WORK EXPERIENCE

Machine Learning Engineer

Skypler AI, Chennai, Tamil Nadu

Oct 2024 – Nov 2025

- Architected an AI-powered multi-agent Python framework that automatically transforms design inputs into production-ready frontend interfaces and backend code (schemas, APIs, services) using LLM-driven planning and synthesis, reducing engineering setup time by 75%.
- Led fine-tuning and knowledge-distillation pipelines for parameter-efficient code models on large-scale custom code datasets, applying structured pruning techniques to achieve 30% faster inference with minimal accuracy loss; deployed distilled models for lightweight code-generation workloads.
- Refactored and stabilized core agents to robustly handle typed errors, dynamic generation plans, and multi-step execution flows, increasing end-to-end build success rate from 60% to 85% across diverse application workflows.
- Collaborated on full MLOps pipelines (SFT → Pruning → Distillation), optimizing for constrained GPU environments and resolving performance, CUDA, and configuration issues.

Data Scientist cum Machine Learning Engineer Intern

Skypler AI, Chennai, Tamil Nadu

Aug 2024 – Oct 2024

- Built real-time video analytics system (object detection, face recognition, ReID) across multiple CCTV feeds, improving accuracy by 30% and reducing inference latency by 40%.
- Developed a production-ready vehicle monitoring system with Automated License Plate Recognition (ALPR) and weight extraction using image-processing pipelines, achieving over 92% license-plate recognition accuracy.
- Optimized ML models using quantization, distillation, and ROI-based inference, improving edge-device throughput from 8 FPS to 20+ FPS.
- Improved tracking reliability and environmental robustness (low light, motion blur), reducing false-positive rates by 35%.

Machine Learning Engineer Intern

NEVAR Systems, Puducherry

Feb 2024 - Apr 2024

- Built a high-accuracy crack detection system using YOLO and Roboflow-annotated datasets, improving defect localization precision by 35% across diverse structural surfaces.
- Integrated SAM (Segment Anything Model) for pixel-level crack segmentation, enabling precise boundary extraction and reducing measurement error by 40%.
- Enhanced an end-to-end automated inspection pipeline that detects, segments, and quantifies crack dimensions (length, width, area), improving maintenance decision speed by 50%.
- Optimized model training through data augmentation, hyperparameter tuning, Roboflow dataset versioning, and improved preprocessing/threshold calibration, increasing YOLO crack-detection accuracy while significantly reducing false positives.

PROJECTS

- **Crack Detection Tool**, Enhanced a crack detection tool with auto-image annotation using YOLOv8 and Segment Anything Model, improving anomaly detection and accuracy. [GitHub](#)
- **ASL Detection with Deep Learning**, Devised an ASL gesture recognition system utilizing Convolutional Neural Networks (CNNs). Achieved accurate recognition of gestures [GitHub](#)
- **Language translation using KerasNLP**, Orchestrated a language translation tool using KerasNLP, enhancing text translation accuracy between English and regional languages, achieving high translation quality with real-time data input. [GitHub](#)

CERTIFICATES

- **Machine Learning & Deep Learning in Python & R** [Certificate](#)
- **Microsoft Machine Learning Challenge** [Microsoft Portal](#)
- **Convolutional Neural Networks:Deep Learning** [Certificate](#)
- **Data Science Mastery with Python** [Certificate](#)